

Supporting Information:
A thermally formed flavin radical pair that forms fast enough to
matter cannot sense the geomagnetic field

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S1 Software and environment

Spin dynamics: `qbscreen` (this work), exact Liouville-space solution with NumPy. Electronic structure: PySCF 2.12.1 (B3LYP, def2-SVP and def2-TZVP, RI-J density fitting); geometries: xtb 6.7.1 (GFN2-xTB, L-BFGS optimizer, tight convergence). Every number in the main text and in the tables below is produced by the scripts `qbscreen/calibration_run.py`, `qbscreen/calibration_extras.py`, `qbscreen/make_numbers_tex.py` and `qbscreen/make_si_tables.py` from the stored results in `calibration_res`.

S2 Electronic coupling scan

Methylamine was placed on the si face of lumiflavin with its nitrogen on the ring normal through N5 at distance r and its lone pair directed at N5. Twist rotates the amine about that axis; tilt tips the axis off the normal. Geometries whose closest intermolecular contact is below 2.4 Å are steric clashes of the rigid placement and were excluded from the fits (Table S1).

S3 Inner-sphere reorganization energy

Four-point energies (hartree) of each redox couple, state s at the optimized geometry G of state s' , and the resulting half contributions $\lambda_i/2 = \frac{1}{2}\{[E_0(G_1) - E_0(G_0)] + [E_1(G_0) - E_1(G_1)]\}$ (Table S2). State 0 is the closed-shell species, state 1 the radical ion.

S4 Hyperfine couplings

Isotropic Fermi-contact couplings from the UKS spin density at each nucleus (B3LYP/def2-TZVP), six largest per radical (Table S3). The flavin anion N5 and N10 values, 441 and 165 μT , compare with 523 and 189 μT of Lee et al.; the spin model uses the latter. The large aminium couplings come from $\alpha\text{-C-H}$ protons hyperconjugated with the nitrogen-centred orbital and depend on conformation.

S5 Catalytic-competence bound: full grid

Table S4: Upper bound on Δ for thermal formation at k_{cat} ,
with the lifetime and exchange coupling at the bound.

t (meV)	λ (eV)	k_{cat} (s ⁻¹)	Δ_{max} (eV)	τ (s)	$ J $ (MHz)
22	0.7	1	0.80	2.3×10^{-13}	2.9×10^5
22	0.7	10	0.74	2.1×10^{-13}	3.2×10^5
22	0.7	100	0.68	2.1×10^{-13}	3.5×10^5
22	1.0	1	0.78	3.5×10^{-13}	3.0×10^5
22	1.0	10	0.71	4.9×10^{-13}	3.3×10^5
22	1.0	100	0.64	7.7×10^{-13}	3.7×10^5
22	1.4	1	0.70	6.4×10^{-12}	3.3×10^5
22	1.4	10	0.62	1.5×10^{-11}	3.8×10^5
22	1.4	100	0.53	3.9×10^{-11}	4.4×10^5
109	0.7	1	0.87	1.6×10^{-13}	6.6×10^6
109	0.7	10	0.82	1.3×10^{-13}	7.0×10^6
109	0.7	100	0.76	1.1×10^{-13}	7.6×10^6
109	1.0	1	0.88	1.2×10^{-13}	6.6×10^6
109	1.0	10	0.81	1.5×10^{-13}	7.1×10^6
109	1.0	100	0.74	2.0×10^{-13}	7.8×10^6
109	1.4	1	0.81	1.1×10^{-12}	7.1×10^6
109	1.4	10	0.73	2.1×10^{-12}	7.8×10^6
109	1.4	100	0.65	4.5×10^{-12}	8.8×10^6
160	0.7	1	0.89	1.7×10^{-13}	1.4×10^7
160	0.7	10	0.84	1.3×10^{-13}	1.5×10^7
160	0.7	100	0.78	1.1×10^{-13}	1.6×10^7
160	1.0	1	0.90	1.1×10^{-13}	1.4×10^7
160	1.0	10	0.83	1.3×10^{-13}	1.5×10^7
160	1.0	100	0.76	1.7×10^{-13}	1.6×10^7
160	1.4	1	0.84	8.4×10^{-13}	1.5×10^7
160	1.4	10	0.76	1.6×10^{-12}	1.6×10^7
160	1.4	100	0.68	3.3×10^{-12}	1.8×10^7
290	0.7	1	0.92	1.9×10^{-13}	4.4×10^7
290	0.7	10	0.86	1.4×10^{-13}	4.7×10^7
290	0.7	100	0.81	1.2×10^{-13}	5.0×10^7
290	1.0	1	0.93	1.1×10^{-13}	4.4×10^7
290	1.0	10	0.87	1.2×10^{-13}	4.7×10^7
290	1.0	100	0.80	1.5×10^{-13}	5.1×10^7
290	1.4	1	0.88	6.2×10^{-13}	4.6×10^7
290	1.4	10	0.80	1.1×10^{-12}	5.1×10^7
290	1.4	100	0.72	2.2×10^{-12}	5.6×10^7

S6 Magnetic field effect along the coupling-locked locus

For the MAO contact pair ($r = 0.35$ nm, point-dipole $D = -1817$ MHz), the exchange coupling and lifetime follow from t through equations 2–4 of the main text, with the adiabatic interpolation. MFE in % at 50 μ T, coherent and with $T_2^e = 1$ μ s, field at 0 and 90° to the interradsical axis.

S7 Partner hyperfine coupling and field orientation

Coherent MFE (%) of the MAO contact pair with $J = 0$ and long lifetimes, as a function of the largest partner hyperfine coupling, with the point-dipole D . This is the most favourable case the contact dipolar coupling allows; it ignores the competence bound. The same bound with the spin-distributed D is in Table S9.

Table S1: Fragment-projection coupling t between the flavin LUMO and the amine HOMO (B3LYP/def2-SVP).

r (Å)	twist (°)	tilt (°)	t (meV)	closest contact (Å)	used
3.00	0	-15	284.9	2.84	yes
3.00	0	0	615.4	2.50	yes
3.00	0	15	882.7	1.99	no (clash)
3.00	60	0	248.1	2.51	yes
3.25	0	-15	177.3	3.09	yes
3.25	0	0	424.8	2.75	yes
3.25	0	15	623.0	2.18	no (clash)
3.25	60	0	179.3	2.76	yes
3.50	0	-15	109.1	3.34	yes
3.50	0	0	290.4	3.00	yes
3.50	0	15	428.6	2.38	no (clash)
3.50	60	0	129.6	3.01	yes
3.75	0	-15	66.7	3.56	yes
3.75	0	0	197.2	3.25	yes
3.75	0	15	290.5	2.58	yes
3.75	60	0	93.6	3.23	yes
4.00	0	-15	40.9	3.79	yes
4.00	0	0	133.5	3.50	yes
4.00	0	15	195.2	2.80	yes
4.00	60	0	67.7	3.41	yes
4.50	0	-15	16.6	4.26	yes
4.50	0	0	62.6	4.00	yes
4.50	0	15	88.2	3.24	yes
4.50	60	0	36.8	3.81	yes
5.00	0	-15	8.0	4.73	yes
5.00	0	0	31.0	4.50	yes
5.00	0	15	42.7	3.70	yes
5.00	60	0	21.2	4.23	yes
5.50	0	-15	4.0	5.21	yes
5.50	0	0	15.1	5.00	yes
5.50	0	15	21.9	4.17	yes
5.50	60	0	11.5	4.66	yes
6.00	0	-15	1.8	5.69	yes
6.00	0	0	6.5	5.50	yes
6.00	0	15	10.5	4.64	yes
6.00	60	0	5.3	5.10	yes

S8 Dipolar coupling from the spin distribution

At van der Waals contact the point-dipole approximation concentrates each radical's spin on one point. We summed the dipolar tensor over the Löwdin atomic spin populations of the two radicals (UKS B3LYP/def2-SVP, vertical ions at the contact geometry): $T_{ab} = d \sum_{ij} \rho_i \rho_j (\delta_{ab} - 3\hat{r}_a \hat{r}_b) / r_{ij}^3$, $d = 52.04 \text{ MHz nm}^3$, and took $D = \frac{3}{4} T_n$ from the principal value of largest magnitude, which reproduces $D = -77.9/r^3 \text{ MHz}$ for a single pair of unit spins (Table S8). The resulting bound, with $J = 0$ and long lifetimes, is in Table S9.

S9 Cryptochrome control

FAD $\bullet^-$ -W324 $\bullet^+$ at 1.90 nm, $D = -11.4 \text{ MHz}$, lifetime and T_2^e of 1 μs , exchange from $J(r) = J_0 e^{-\beta r}$ with $\beta = 14 \text{ nm}^{-1}$ over $8 \times 10^{12} \leq |J_0|/\mu\text{T} \leq 8 \times 10^{14}$ and both signs (Efimova and Hore). Spin- $\frac{1}{2}$: N5 and N10 as effective spin- $\frac{1}{2}$ nuclei; spin-1: ^{14}N as $I = 1$. Trp H β 1 is 45 MHz in both. Powder: Gauss-Legendre average over the field direction.

Table S2: Four-point energies (B3LYP/def2-SVP at GFN2-xTB geometries).

species	$E_0(G_0)$	$E_0(G_1)$	$E_1(G_1)$	$E_1(G_0)$	$\lambda_i/2$ (eV)
lumiflavin	-871.480317	-871.473953	-871.539794	-871.531658	0.197
methylamine	-95.780457	-95.771484	-95.461713	-95.439105	0.430
benzylamine	-326.673644	-326.663153	-326.374740	-326.364444	0.283
alanine	-323.509936	-323.476155	-323.178642	-323.158966	0.727

Table S3: DFT isotropic hyperfine couplings.

radical	nucleus	a_{iso} (MHz)	a_{iso} (mT)
flavin $\bullet^-$	H29	+17.9	+0.639
flavin $\bullet^-$	H30	+17.8	+0.636
flavin $\bullet^-$	N4	+12.4	+0.441
flavin $\bullet^-$	H25	+11.3	+0.403
flavin $\bullet^-$	H26	+11.3	+0.402
flavin $\bullet^-$	H22	-10.5	-0.374
benzylaminium $\bullet^+$	H10	+256.9	+9.167
benzylaminium $\bullet^+$	H11	+196.2	+7.001
benzylaminium $\bullet^+$	N0	+24.8	+0.885
benzylaminium $\bullet^+$	H8	-23.0	-0.821
benzylaminium $\bullet^+$	H9	-20.7	-0.737
benzylaminium $\bullet^+$	H14	-12.3	-0.439
methylaminium $\bullet^+$	H2	+363.1	+12.955
methylaminium $\bullet^+$	H6	+82.7	+2.951
methylaminium $\bullet^+$	H5	+81.9	+2.923
methylaminium $\bullet^+$	H3	-59.3	-2.116
methylaminium $\bullet^+$	H4	-59.3	-2.115
methylaminium $\bullet^+$	N1	+30.5	+1.090
alaninium $\bullet^+$	H9	+58.3	+2.080
alaninium $\bullet^+$	H6	+30.8	+1.098
alaninium $\bullet^+$	N2	+18.6	+0.665
alaninium $\bullet^+$	H11	-17.9	-0.639
alaninium $\bullet^+$	H10	-17.7	-0.632
alaninium $\bullet^+$	H7	+14.1	+0.502

Table S5: Inverted region, $\Delta = 1.78$, $\lambda = 0.97$ eV.

t (meV)	$ J $ (MHz)	τ (s)	coh. 0°	coh. 90°	T_2^e 0°	T_2^e 90°
290	2.3×10^7	5.7×10^{-11}	1.1×10^{-14}	0	0	-1.0×10^{-13}
160	7.0×10^6	5.7×10^{-11}	0	2.2×10^{-14}	0	-8.9×10^{-14}
100	2.7×10^6	5.9×10^{-11}	0	0	0	-1.3×10^{-13}
30	2.4×10^5	9.3×10^{-11}	-2.2×10^{-14}	-2.2×10^{-14}	0	-5.6×10^{-13}
22	1.3×10^5	1.3×10^{-10}	-6.7×10^{-14}	-4.4×10^{-14}	-2.2×10^{-14}	-1.3×10^{-12}
10	2.7×10^4	3.9×10^{-10}	-1.6×10^{-11}	-1.8×10^{-11}	-1.6×10^{-11}	-3.5×10^{-11}
3	2.4×10^3	3.8×10^{-9}	-8.7×10^{-8}	-7.4×10^{-7}	-8.7×10^{-8}	-7.3×10^{-7}
1	2.7×10^2	3.4×10^{-8}	-1.5×10^{-5}	3.9×10^{-6}	-1.5×10^{-5}	3.6×10^{-6}
0.3	2.4×10^1	3.7×10^{-7}	-4.3×10^{-5}	6.0×10^{-6}	-4.7×10^{-5}	-2.8×10^{-6}
0.1	2.7×10^0	3.4×10^{-6}	-4.9×10^{-5}	6.8×10^{-6}	-7.2×10^{-5}	-1.3×10^{-4}
0.03	2.4×10^{-1}	3.7×10^{-5}	-5.0×10^{-5}	6.9×10^{-6}	-6.9×10^{-5}	-1.5×10^{-3}
0.01	2.7×10^{-2}	3.4×10^{-4}	-5.0×10^{-5}	6.9×10^{-6}	-2.3×10^{-5}	-9.4×10^{-3}

Table S6: Activationless, $\Delta = \lambda = 1.0$ eV.

t (meV)	$ J $ (MHz)	τ (s)	coh. 0°	coh. 90°	T_2^e 0°	T_2^e 90°
290	4.1×10^7	1.0×10^{-13}	0	0	0	-1.1×10^{-14}
160	1.2×10^7	1.0×10^{-13}	0	0	1.1×10^{-14}	0
100	4.8×10^6	1.1×10^{-13}	0	0	0	0
30	4.4×10^5	1.7×10^{-13}	0	0	0	0
22	2.3×10^5	2.3×10^{-13}	0	0	-1.1×10^{-14}	-1.1×10^{-14}
10	4.8×10^4	7.1×10^{-13}	2.2×10^{-14}	2.2×10^{-14}	0	0
3	4.4×10^3	6.8×10^{-12}	8.3×10^{-12}	9.1×10^{-12}	8.3×10^{-12}	9.1×10^{-12}
1	4.8×10^2	6.1×10^{-11}	1.5×10^{-8}	5.1×10^{-8}	1.5×10^{-8}	5.1×10^{-8}
0.3	4.4×10^1	6.7×10^{-10}	-2.4×10^{-5}	5.1×10^{-6}	-2.4×10^{-5}	5.1×10^{-6}
0.1	4.8×10^0	6.1×10^{-9}	-4.3×10^{-5}	6.9×10^{-6}	-4.3×10^{-5}	6.9×10^{-6}
0.03	4.4×10^{-1}	6.7×10^{-8}	-5.2×10^{-5}	6.9×10^{-6}	-5.2×10^{-5}	6.4×10^{-6}
0.01	4.8×10^{-2}	6.1×10^{-7}	-5.0×10^{-5}	6.6×10^{-6}	-5.4×10^{-5}	-1.2×10^{-5}

Table S7: Dependence on the partner hyperfine coupling.

partner a (MHz)	τ (μ s)	0°	45°	90°
5	1	-2.4×10^{-7}	-1.2×10^{-7}	3.6×10^{-8}
5	100	-2.8×10^{-7}	-1.2×10^{-7}	3.6×10^{-8}
44	1	1.5×10^{-6}	9.4×10^{-7}	2.9×10^{-7}
44	100	1.5×10^{-6}	9.1×10^{-7}	2.8×10^{-7}
100	1	1.5×10^{-6}	4.8×10^{-6}	1.3×10^{-6}
100	100	1.6×10^{-6}	4.8×10^{-6}	1.3×10^{-6}
257	1	-5.0×10^{-5}	-1.8×10^{-5}	6.7×10^{-6}
257	100	-5.0×10^{-5}	-1.8×10^{-5}	6.9×10^{-6}
363	1	-8.8×10^{-5}	-3.7×10^{-5}	1.1×10^{-5}
363	100	-8.8×10^{-5}	-3.7×10^{-5}	1.1×10^{-5}

Table S8: Point-dipole and spin-distributed dipolar couplings at the contact geometry.

r ($\text{\AA}$)	D_{point} (MHz)	centroid dist. (nm)	D_{spin} (MHz)	E (MHz)	principal values (MHz)
3.50	-1817	0.348	-914	-26	-1219, 558, 662
4.00	-1217	0.397	-674	-17	-899, 415, 484

Table S9: Coherent MFE (%) with the spin-distributed D , $J = 0$.

enzyme	partner a (MHz)	τ (μ s)	0°	45°	90°
MAO	5	1	-3.4×10^{-6}	-1.7×10^{-6}	5.6×10^{-7}
MAO	5	10	-3.9×10^{-6}	-1.8×10^{-6}	5.6×10^{-7}
MAO	5	100	-4.0×10^{-6}	-1.8×10^{-6}	5.6×10^{-7}
MAO	44	1	6.2×10^{-6}	1.5×10^{-5}	4.3×10^{-6}
MAO	44	10	6.4×10^{-6}	1.6×10^{-5}	4.2×10^{-6}
MAO	44	100	6.4×10^{-6}	1.6×10^{-5}	4.2×10^{-6}
MAO	100	1	-1.0×10^{-4}	-2.5×10^{-5}	1.7×10^{-5}
MAO	100	10	-1.0×10^{-4}	-2.5×10^{-5}	1.8×10^{-5}
MAO	100	100	-1.0×10^{-4}	-2.5×10^{-5}	1.8×10^{-5}
MAO	257	1	-4.3×10^{-4}	-1.9×10^{-4}	6.4×10^{-5}
MAO	257	10	-4.3×10^{-4}	-1.9×10^{-4}	6.6×10^{-5}
MAO	257	100	-4.3×10^{-4}	-1.9×10^{-4}	6.6×10^{-5}
MAO	363	1	-3.3×10^{-4}	-1.3×10^{-4}	7.7×10^{-5}
MAO	363	10	-3.3×10^{-4}	-1.3×10^{-4}	7.9×10^{-5}
MAO	363	100	-3.3×10^{-4}	-1.3×10^{-4}	7.9×10^{-5}
DAO	5	1	-1.1×10^{-5}	-5.4×10^{-6}	1.9×10^{-6}
DAO	5	10	-1.3×10^{-5}	-5.7×10^{-6}	1.9×10^{-6}
DAO	5	100	-1.3×10^{-5}	-5.7×10^{-6}	1.9×10^{-6}
DAO	58	1	-8.6×10^{-5}	3.5×10^{-6}	2.2×10^{-5}
DAO	58	10	-8.6×10^{-5}	4.3×10^{-6}	2.2×10^{-5}
DAO	58	100	-8.6×10^{-5}	4.3×10^{-6}	2.2×10^{-5}
DAO	100	1	-4.1×10^{-4}	-1.5×10^{-4}	5.3×10^{-5}
DAO	100	10	-4.1×10^{-4}	-1.5×10^{-4}	5.5×10^{-5}
DAO	100	100	-4.1×10^{-4}	-1.5×10^{-4}	5.5×10^{-5}

Table S10: Cryptochrome MFE (%) at 50μ T.

model	$ J_0 $ (μ T)	J (MHz)	0°	90°	powder
spin half	8×10^{12}	-1.26	-0.236	-1.742	-1.422
spin one N	8×10^{12}	-1.26	+0.186	-1.447	—
spin half	8×10^{12}	+1.26	+0.199	-1.048	-0.880
spin one N	8×10^{12}	+1.26	+0.262	-1.285	—
spin half	8×10^{13}	-12.57	-0.025	-0.720	-0.501
spin one N	8×10^{13}	-12.57	-0.016	-1.436	—
spin half	8×10^{13}	+12.57	+0.058	-0.284	-0.177
spin one N	8×10^{13}	+12.57	+0.029	-0.206	—
spin half	8×10^{14}	-125.73	-0.012	-0.322	-0.254
spin one N	8×10^{14}	-125.73	-0.037	-0.222	—
spin half	8×10^{14}	+125.73	-0.021	-0.326	-0.222
spin one N	8×10^{14}	+125.73	-0.022	-0.234	—